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PI Name:	Rithidech, Kanokporn Ph.D.		
Project Title:	Dose-rate effects of protons on the induction of genomic instability in vivo		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Comments:			
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Anderson, Carl (BNL) Simon, Sanford (Stony Brook University) Whorton, Elbert (University of Texas Medical Branch at Galveston)		
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The radiation component of space environment is very complex. In addition to galactic cosmic rays (GCRs), radiation produced during solar particle events (SPEs) is one of the major radiation sources during a space mission to the Moon and Mars. The occurrence of SPEs, with an approximately 11-year solar cycle, is unpredictable and each SPE varies in size and spectral characteristics, including the duration of event (which can last from several hours to a few days). While the biological effects of GCR particles, in particular those with large mass (Z) and high energy (E) (known as HZE particles, such as 56Fe ions) have been investigated extensively, little is known about the biological effects of SPE components. We conducted a series of studies to fill this knowledge gap.
A large SPE can release a very high flux of charged-particle radiation of which approximately (98%) are protons with energy less than 150 MeV. For these reasons, protons with the energy of less than 150 MeV possess a major threat to astronauts during a space mission, in particular throughout the extra-vehicular activity (EVA). Additionally, protons with this range of energy present serious hazards to the crew of the International Space Station (ISS). Hence, health risks from exposure to protons are one of the major concerns for exploration missions if the astronauts are improperly protected. It is therefore essential to improve our understanding of the potential biological effects of exposure to protons for a better assessment of health risks.

As the first step in investigating the in vivo biological effects of protons with energy less than 150 MeV, we initiated a series of studies focusing on 100 MeV protons. We determined the type, the frequency, and the distribution of initial and subsequent chromosome aberrations (CAs), and the expression of genomic instability (assessed as late occurring chromosomal damage) induced in vivo in bone

Task Description:	marrow (BM) cells of mice exposed to 100 MeV protons at the dose and dose rates encountered in space. We are the first to determine the efficiency of 100 MeV protons in inducing genomic instability, the phenomenon known to be highly associated with cancer development. Further, our studies were designed to improve our understanding of the molecular basis for dose-rate dependencies of proton-induced genomic instability in vivo that, in turn, should help reduce the uncertainties in assessing the risk of space flight. To do this, we simultaneously studied: • the frequencies of early- and late-occurring chromosome aberrations (CAs), by means of the whole mouse genome multi-color fluorescence in situ hybridization (mFISH), • the kinetics of repair of the damage, determined by the formation and reduction of phosphorylated histone 2AX (gamma H2AX) which is an important biomarker of double strand breaks, • the kinetics of nuclear factor-kappa B (NF-kappa B) activation, • the expression of selected NF-kappa B target proteins known to be involved in inflammatory response, i.e. pro-inflammatory cytokines. Of note, these biological endpoints were measured in BM cells collected from the same mouse. Hence, our research is the first to use integrative in vivo endpoints in the tissue (BM) of the same mouse that is the target of radiation-induced myeloid leukemia to evaluate the biological effects of protons produced during SPEs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The resulting data deliver information that is of high priority for NASA. This is because the data will not only contribute to reducing the uncertainties in the assessment of cancer risk from exposure to protons during space flights but will also improve the understanding of the molecular basis for dose/dose-rate dependencies of proton-induced genomic instability and chronic inflammation. Hence, the obtained data are critically important for understanding risks associated with space travel as well as developing advanced proton therapies for cancer treatment on earth. Further, knowing the pathways associated with response to radiation exposure could be useful for planning countermeasure strategies to protect astronauts or space travelers in the space environment. Likewise, such knowledge can also have Earth benefits because a large number of people on Earth are exposed to radiation at varying doses and in many different ways. These include: individuals exposed to low levels of radiation in daily life (e.g. medical diagnosis or airport safety) or an unavoidable exposure to low-level background radiation; patients who receive high doses of radiation from radiation therapy; victims of nuclear terrorism; members of the armed forces; workers in the nuclear power industry; and people living in homes surrounding the power plants.
Task Progress:	We have completed the project. We exposed mice to 0, 0.5, or 1.0 Gy of 100 MeV protons (delivered at 0.5 or 1 cGy/min). Bone marrow cells were collected from groups of mice (5 mice in each treatment group) at various times post-irradiation (up to 6 months post-irradiation). Various endpoints, as shown below, were used for determining the impacts of dose-rate effects after exposure to 100 MeV protons: • The analysis of chromosome aberrations, by using a whole genome mFISH analysis • The analysis of gamma H2AX, by means of flow cytometry • The analysis of inflammatory responses, by using the Enzyme-Linked Immunosorbance Assay (ELISA) to determine the levels of activated NF-kappa B and related pro-inflammatory cytokines Highly significant numbers of abnormal cells and CAs related to those in non-irradiated controls were detected in BM cells collected from exposed mice at each time point included in the study, regardless of the dose rate of 100 MeV protons. Such findings indicated that both 0.5 and 1.0 Gy of 100 MeV protons, delivered at 0.5 or 1 cGy/min, are capable of inducing genomic instability in BM cells (the target cell for radiation induced myeloid leukemia). Further, the resulting data indicated a slightly higher number of abnormal cells and chromatid breaks in BM cells of exposed mice when a dose rate of 1.0 cGy/min) was used, as compared to that with a dose rate of 0.5 cGy/min. Although these increases were not statistically significant, our data provide evidence of more energy from a higher dose rate deposited leading to a higher level of chromosomal damage but lower ability to repair the damage. Of note, similar findings were observed, relating to the levels of activated NF-kappa B and relevant pro-inflammatory cytokines [Rithidech et al. REB, 2010, 85(5):432-447]. We also found that a single dose of 0.5 Gy or 1 Gy of 100 MeV protons induced significantly high levels of activated NF-kappa B and pro-inflammatory cytokines in BM cells from irradiated mice, related to the corresponding non-irradiated control levels, at early and late time-points, regardless of the dose and dose-rate of proton-irradiation. These results indicated the persistence of NF-kB activation and expression of pro-inflammatory cytokines in BM cells after exposure of mice to 0.5 or 1.0 Gy of 100 MeV protons, delivered at the dose rates occurring in space. Our new set of in vivo data suggested higher levels of NF-kB activation and expression of pro-inflammatory cytokines with the higher dose rate (1 cGy/min), although there were no significant differences in the levels of these proteins between the two dose rates used in this study. Our data provide important information for space flight missions because these findings suggest the occurrence of chronic inflammation in exposed mice that, in turn, may have an important impact on health outcome during space flights. The results also offer opportunities for developing new biological countermeasures (i.e. anti-inflammatory agents) to mitigate radiation-induced deleterious effects. In summary, the highlights of the resulting data obtained from our project are: • A single dose of 0.5 or 1.0 Gy of 100 MeV protons delivered at 0.5 and 1 cGy/min (the dose and dose rates similar to what the astronauts encounter in space) is capable of inducing genomic instability and chronic inflammation in BM cells of exposed mice. • These findings are important because it has been well recognized that genomic instability is a key event in carcinogenesis and that there is a link between chronic inflammation and cancer induction. • There was a trend of higher effects when a higher dose rate was used.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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